For Sigmoid Function

$$f(x) = \text{Sigmoid}(x) = \frac{1}{1+e^{-x}}$$

Remember:

$$\frac{\partial}{\partial x}\left(\frac{1}{x}\right) = -\frac{1}{x^2}$$

$$f'(x) = \frac{e^{-x}}{(1+e^{-x})(1+e^{-x})} = \frac{1*(1+e^{-x}-1)}{(1+e^{-x})(1+e^{-x})}$$

Sigmoid (aka Logistic function)

It is a *squashing* function used to convert an unbounded values to a bounded probabilistic space *(from 0 to 1)*

$$f'(x) = \frac{1}{(1+e^{-x})}\left|\frac{(1+e^{-x}-1)}{(1+e^{-x})}\right|$$

$$= \frac{1}{(1+e^{-x})}\left|\frac{(1+e^{-x})}{(1+e^{-x})}\ \frac{-1}{(1+e^{-x})}\right|$$

$$\Rightarrow\ f'(x) = f(x)\,[1-f(x)]$$

$$f'(x) = \frac{1}{(1+e^{-x})}\left|1-\frac{1}{(1+e^{-x})}\right|$$